

Reports from Other Journals

News from Online—Plants and Photosynthesis

by Lynn M. Diener

Science Department, Mount Mary College, Milwaukee, Wisconsin 53222
dienerl@mtmary.edu

Chemists everywhere will soon be celebrating Earth Day. Are you among them? This year's theme for Earth Day is Plants: The Green Machines. Plants are at the base of the energy pyramid. They are critically important because they fix carbon dioxide into carbohydrates that can be utilized by other organisms (like us!) for the production of cellular energy. Plants also produce as a byproduct all of the oxygen we breathe. Your students are all familiar with plants, but are they all familiar with photosynthesis (see Figure 1) and the chemistry that plants routinely perform? This article explores Web sites that highlight the chemistry of plants.

Photosynthesis Basics

Instead of ingesting carbon like humans do, plants get their carbon from the air in the form of carbon dioxide (CO_2). Many Web sites summarize photosynthesis, including its chemical reactions, enzymes, photosystems, and the light and dark reactions. I want to share a couple of sites that are particularly interesting. For a thorough but easy-to-read overview, this article by Wim Vermaas of Arizona State University is a good choice (1). It talks about some applications of photosynthesis, specifically about genetic modification with an eye toward improving plant productivity. Another overview can be found at Estrella Mountain Community College by Mike Farabee (2). The strength of this Web site is its great illustrations of chlorophyll. It has a number of other illustrations that I like as well, including an absorption spectrum of different plant pigments. A photosynthesis Web site by June B. Steinberg of National-Louis University provides an overview of the subject along with a series of animations to illustrate the concepts (3). These animations make the difference between cyclic and noncyclic photophosphorylation very clear and easy to understand.

For elementary and middle school students (although high school students would probably also enjoy it), I recommend the "Illuminating Photosynthesis" flash interactive presentation from *NOVA Online* (4). This simple yet informative program lets students see the connection between plants, light, water, and carbon dioxide in a fun and interactive platform. The puzzlers (three trivia questions) are informative and also challenging enough to lead to a few head-scratches.

If your students enjoy videos and video clips, the Teachers' Domain hosts a couple of excellent choices on this topic. The first is a simple introduction to photosynthesis and the way that plants get their food. The video clip titled "Photosynthesis" is from *NOVA: Earth* (5). The second video highlights the impact of the first photosynthetic organisms on Earth as they filled our planet's atmosphere with oxygen over billions of years (6). Your

students might be surprised to learn just how instrumental plants and other photosynthetic organisms were to the establishment of our oxygen-rich atmosphere. With a free account from the Teachers' Domain, you gain access to all of the videos and resources they host. The last video, which is from *Newton's Apple*, gives an overview of photosynthesis by answering questions from viewers (7). This video talks straightforwardly about the chemistry and biology of photosynthesis.

For a fun and affordable laboratory activity, I recommend "The response of leaf discs from sun and shade plants to green light" found at the Science and Plants for Schools Web site (8). Leaf discs are submerged in CO_2 -rich water (thanks to addition of sodium bicarbonate), and the discs are exposed to white and green light. If the discs are able to photosynthesize, they produce oxygen as a byproduct. This, in turn, allows the discs to float. It is a clever activity that requires students to understand a little of the chemistry of photosynthesis as well as the importance of light to this process.

Chloroplasts

In the previous section, we learned about the process of photosynthesis, but I think it helps students to see where in the cell the process actually takes place. Biology4Kids provides an overview that breaks down the chloroplast, highlighting the different parts from the various membranes to the thylakoids (9). The site also teaches students what takes place in the different parts of the chloroplast.

The University of Illinois at Urbana-Champaign hosts Virtual Cell (10). The virtual cell is a single plant cell, and the program allows you to slice the cell open, zoom in, and even dissect organelles inside the cell. I like to zoom in on the chloroplast and cut it open to see its inner workings. There is text to go along with each organelle and every zoom you do into the virtual cell.

The McGraw-Hill Online Learning Center has a visual chloroplast quiz available (11). After going through the Biology4Kids and the University of Illinois Web sites, your students should have no trouble identifying the thylakoid and the inner and outer membranes in the interactive diagram.

Plant Pigments

The first pigment that typically comes to mind when thinking about plants is chlorophyll. Many of your students have heard of it, and they might know that chlorophyll is found within the chloroplasts and gives plants their green color. Let us start by learning a little more about chlorophyll; then we can learn about some of the other plant pigments.



Figure 1. Artistic representations of photosynthesis. (Left panel) Evocation of photosynthesis focusing on trees as major contributors. Through photosynthesis, trees create oxygen, food, wood, medicine, shade, a habitat for animals, and creative inspiration. (Middle panel) Painting of a microscopic cross-section of a leaf, describing the complex structure and dynamic processes of the photosynthesizing factory. (Right panel) Depiction of chlorophyll and its molecular structure, showing the shape of the molecule and its interaction with sunlight to give plants their beautiful green quality. All reproduced with permission of the artists (left to right): Mishell Sommer-Bayer; Alyssa Olson; and Chelsea Hassi.¹

Bassam Z. Shakhshiri hosts the Science Is Fun Web site from the University of Wisconsin—Madison. At this site you can find a writeup of chlorophyll when it was chosen as the chemical of the week (12). One page on the site highlights the structure of chlorophyll. The site also discusses heme, a compound found in blood and one that plays roles in oxidation—reduction reactions in living organisms.

For a bit of history about chlorophyll, head over to the University of Bristol's School of Chemistry (13). Paul May has put together a concise and intriguing history that begins with the finding that plants produce oxygen. It ends with the discovery of the equation for production of oxygen by plants and chlorophyll's role in that equation.

Web Exhibits, which is an online interactive museum, has an easily accessible exhibit on chlorophyll (14). It focuses on various chlorophyll-related topics, including what you see when you look at chlorophyll, how other animals see chlorophyll, and the function of chlorophyll in a plant.

Anyone who has seen fall colors on a beautiful October day knows that chlorophyll is not the only pigment found in plants. An engaging way to introduce these other pigments can be found on the Web site of the publisher Pearson. This site features an interactive lab activity in which students watch a chromatographic separation of the pigments found in a plant leaf and then answer questions about it (15). The activity is interactive and informative and introduces the idea that chlorophyll a and b are not the only pigments found in leaves. The second part of the lab activity focuses on photosynthesis. If you want your students to actually do a similar experiment in the classroom, I recommend the following activity found at APS Education Online (16). This activity has students separate plant pigments, create a standard curve of the different pigments, and then observe the degradation of these pigments over time.

Shakhshiri's Science Is Fun Web site also has an offering that fits the bill when you are discussing other plant pigments (17). The site gives a thorough overview of the various pigments. It also explains why the colors you see in autumn are not visible during the spring and summer months. It is enlightening and beautiful, with a number of pictures of trees

with foliage turning colors. Finally, I recommend a flash animation from GF Awesome, a resource aimed at a middle school audience (18). This interactive animation is actually a complete overview of trees, talking about everything from conifers to deciduous trees and from cellulose to chlorophyll. The beginning of the animation, however, spends a good deal of time on other plant pigments, which is why I highlight it in this section.

Redox Chemistry

Photosynthesis is an easy way to get students excited about oxidation—reduction chemistry. Your students understand plants, and telling them that plants undergo redox reactions can pique their interest in the topic.

Asato at Leeward Community College has an oxidation—reduction resource for students on his Web site (19). It covers the basics of redox chemistry, some extended concepts like electronegativity and types of redox reactions, and exercises that test students' knowledge of the concepts they've just learned. Most relevant to this topic are the everyday examples of redox chemistry. Asato's Web site explores redox topics ranging from metabolism and batteries to photosynthesis. It goes into detail about the redox chemistry of photosynthesis, showing students just how relevant redox chemistry is to their lives.

This *Journal* also has a wonderful resource for teaching redox chemistry: electrochemistry Netorials (20). You can find them in the *JCE* software package, available by subscription for a nominal fee. A series of Netorials teach everything from chemical reactions and stoichiometry to acids and bases. Within the electrochemistry unit are seven Netorials to peruse, covering everything from the basics of redox chemistry to batteries and electrolytic cells. All of the Netorials are organized in a similar fashion, with short portions of informative text, interactive animations, quick student information checkups within the tutorial, and end-of-chapter quizzes for students to continue testing their knowledge.

I hope that you have found something you can use in your class this Earth Day. It is a great opportunity to get your students excited about Earth and conserving its resources. It is also a great

opportunity to teach your students about some relevant and everyday chemistry of plants, the green machines that keep life running on our planet.

Note

1. The artwork in this piece is the result of an ongoing collaboration between Lynn Diener and Mount Mary Art Professor Jordan Acker Anderson, whose art students created and described the graphic art in this article.

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